Figure 6.3 BlindAuction contract diagram with event definitions included

The design of the blind auction smart contract now features three events:

- `AuctionEnded` to announce the end of the auction
- `BiddingStarted` to announce the `Bidding` phase of the auction
- `RevealStarted` to announce the `Reveal` phase of the auction

Let's look at the smart contract code with the addition of these events and their triggers. The entire code (BlindAuction.sol) with events is available in the codebase for this chapter. You can download it into Remix to follow along with the discussion; the following listing shows only the parts that are relevant to events.

Listing 6.1 Blind auction with events (BlindAuction.sol)

```
pragma solidity >=0.4.22 <=0.6.0;
contract BlindAuction {
    // Data types
    ...
    // Enum-uint mapping:
    // Init - 0; Bidding - 1; Reveal - 2; Done - 3                ❶
    enum Phase {Init, Bidding, Reveal, Done}
    ...
    Phase public currentPhase = Phase.Init;

     // Events
    event AuctionEnded(address winner, uint highestBid);          ❷
    event BiddingStarted();                                       ❷
    event RevealStarted ();                                       ❷

    // Modifiers
    modifier validPhase(Phase phase) { ... }

    modifier onlyBeneficiary() { ... }

    constructor() public {
        beneficiary = msg.sender;
    }
```